

Ayush Bhagwat

Backend Engineer (Go) · AI agents · trading systems

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Summary

Backend engineer (Go). At Radisys I worked on the Go control plane that provisions bare-metal servers into Kubernetes clusters for Deutsche Telekom's ~900-site broadband rollout. Now building backends where correctness is the hard part: an AI support agent whose backend, not the model, enforces the rules, and a perpetual-futures exchange. Seeking backend and backend-AI roles (SDE-1/SDE-2), remote or relocation.

Experience

Software Engineer, Backend

Radisys | Infrastructure and Edge Cloud Compute teams, Deutsche Telekom Access 4.0

Jan 2023 – Feb 2024

Bengaluru, India

- Worked on the Go control plane for Deutsche Telekom's ~900-site FTTH rollout, on the team that built the **Pod Bring-up Service (PBS)**: a stateless **Go/Gin microservice** that replaced a legacy Python controller and provisions bare-metal servers into Rancher-managed **K3s clusters** (DNS/DHCP, IPMI PXE boot, SUSE OS install, K3s via Salt, Rancher registration).
- Extended the **notification proxy mock** (Go/Gin) with node-level notification endpoints, so per-server bring-up progress could be verified in **CI** without the live notification system.
- Resolved **19+ bugs** across pod bring-up, installation/deployment and graceful shutdown in a distributed microservice platform; reviewed code and wrote the team's Confluence docs for those flows.
- Traced PBS's provisioning code end to end into **6 sequence diagrams** (server install, DNS/DHCP, OS profiles, K8s install status, kubeconfig), used by the team as the reference for the flow.
- Deployed and ran **Terraform**-provisioned virtual PODs (the full stack on VMs), so squads without physical hardware could test end to end.

Go, Gin, MongoDB, Kubernetes (K3s), Rancher, Helm, SUSE Manager, IPMI, GitLab CI, Terraform, Linux

Projects

AI Customer Support Agent

github.com/MidNight91119/ai-support-agent

Go · LLM tool calling · concurrency

- An AI agent works each ticket by calling backend tools (order lookup, policy search, stock check), then resolves it, asks for details, waits for approval, or escalates to a human.
- **The backend enforces the rules, not the model**: customers touch only their own orders, refunds above a limit need human approval, each order gets at most one refund or replacement, and every argument the model writes is validated before it runs.
- **Safe under concurrent requests**: state changes happen in one short locked section that never waits on the model, so no order is refunded twice; if the AI provider is down, the ticket fails safely with no side effects.

Perpetual Futures Exchange

github.com/MidNight91119/perps-exchange

Go · order book · Binance feed

- **Matching engine with price-time priority**: limit and market orders, best price then oldest first, partial fills, post-only.
- **Margin and leverage**: locks collateral before trading, tracks positions and PnL, and **liquidates on mark price**, not last trade.
- **Funding payments** that tie the contract's price to spot, using a live index price from Binance.

SimpleBank

github.com/MidNight91119/simplebank

Go · gRPC · PostgreSQL · Redis · AWS EKS

- Payments backend with **deadlock-safe concurrent transfers**: one database transaction per transfer, rows locked in a fixed order, every money movement recorded in a ledger.
- Same API over **gRPC and REST**, PASETO/JWT auth with roles, a Redis-queued email worker; CI against real Postgres, and every push to main deploys a Docker image to **AWS EKS**.

Skills

Languages: Go, SQL, TypeScript, Python

Backend: Microservices, distributed systems, concurrency, REST, gRPC, Gin, PostgreSQL, sqlc, MongoDB, Redis

AI: LLM tool calling, agent loops, OpenAI-compatible APIs

Infrastructure: Docker, Kubernetes, Helm, Terraform, AWS (EKS, ECR), Linux, GitHub Actions, GitLab CI

Testing: Unit and integration tests (Go testing, gomock), race detector, black-box API contract tests

Education

B.Tech, Information Technology | CGPA 8.27

Indian Institute of Information Technology, Lucknow (IIIT Lucknow)

2019 – 2023

Lucknow, India

Achievements

- **1000+ problems solved** on Codeforces, AtCoder and LeetCode | **1st place**, Hack-O-Heist hackathon, IIIT Lucknow | **Best Presentation Award**, Radisys 2023.
- VP Membership, Toastmasters Radisys (2023) | E-Cell Coordinator, IIIT Lucknow (2022) | Hult Prize Campus Director (2021).